



LECTURE #14

Image Classification (Continuation)

OBJECTIVE

- EXPLORATORY CLASSIFICATION USING K-MEANS.

STEPS

1. SUPERVISED X SEMI-AUTOMATIC CLASSIFICATION.
2. TYPES OF IMAGE CLASSIFICATION.
3. 4. UNSUPERVISED CLASSIFICATION WITH SCP PLUG-IN.

1 – SUPERVISED CLASSIFICATIONS

WHAT IS SUPERVISED CLASSIFICATION?

PIXEL CLASSIFICATION BASED ON TRAINING SAMPLES PROVIDED BY THE USER.

WHEN TO USE IT

- WHEN YOU KNOW YOUR CLASSES
- WHEN HIGH ACCURACY IS NEEDED
- WHEN THE STUDY AREA HAS WELL-DEFINED LAND-COVER TYPES
-

COMMON ALGORITHMS IN SCP

- **MAXIMUM LIKELIHOOD (WE'LL USE THIS ONE)**
- MINIMUM DISTANCE
- SPECTRAL ANGLE MAPPING
- MLP
- SVM

STEPS IN SCP

1. BUID A BAND STACK

- A. LOAD BANDS;
- B. BAND RESAMPLE;
- C. BAND SET MANAGER (CREATE BAND STACK);

2. CREATE ROIS FOR EACH CLASS

- A. SELECT ALGORITHM
- B. SELECT BAND SET

3. RUN CLASSIFICATION

STEP 1: BUILD A BAND STACK

C. BAND SET MANAGER (CREATE BAND STACK)

MORE DETAILS IN UNSUPERVISED CLASSIFICATION SLIDES.



AFTER CREATING THE NEW BAND SET, CLICK HERE TO VISUALIZE

MENU → SCP → BAND SET

Band set definition

Active band set 1

Date NaT

Root directory

	Band name	Center wavelength	Multiplicative Factor	Additive Factor	Wavelength unit	
1	B2_10M	490.0	1.0	0.0	nm (1 E-9m)	C
2	B3_10M	560.0	1.0	0.0	nm (1 E-9m)	C
3	B4_10M	665.0	1.0	0.0	nm (1 E-9m)	C
4	B5_10M_resampled	705.0	1.0	0.0	nm (1 E-9m)	C
5	B6_10M_resampled	740.0	1.0	0.0	nm (1 E-9m)	C
6	B7_10M_resampled	783.0	1.0	0.0	nm (1 E-9m)	C
7	B8_10M	842.0	1.0	0.0	nm (1 E-9m)	C
8	B8A_10M_resampled	865.0	1.0	0.0	nm (1 E-9m)	C
9	B11_10M_resampled	1610.0	1.0	0.0	nm (1 E-9m)	C
10	B12_10M_resampled	2190.0	1.0	0.0	nm (1 E-9m)	C

DOUBLE-CLICK EACH CELL TO EDIT.

WAVELENGTH UNIT → NM (1E-9 M)

Band quick settings

Wavelength Band order

Wavelength unit nm (1 E-9m)

Date 2020-01-01

Band set tools

☒ Create virtual raster of band set

☐ Create raster of band set (stack bands)

☐ Build band overviews

☐ Band calc expressions

RUN

CREATE VIRTUAL RASTER

SAVE BAND SET

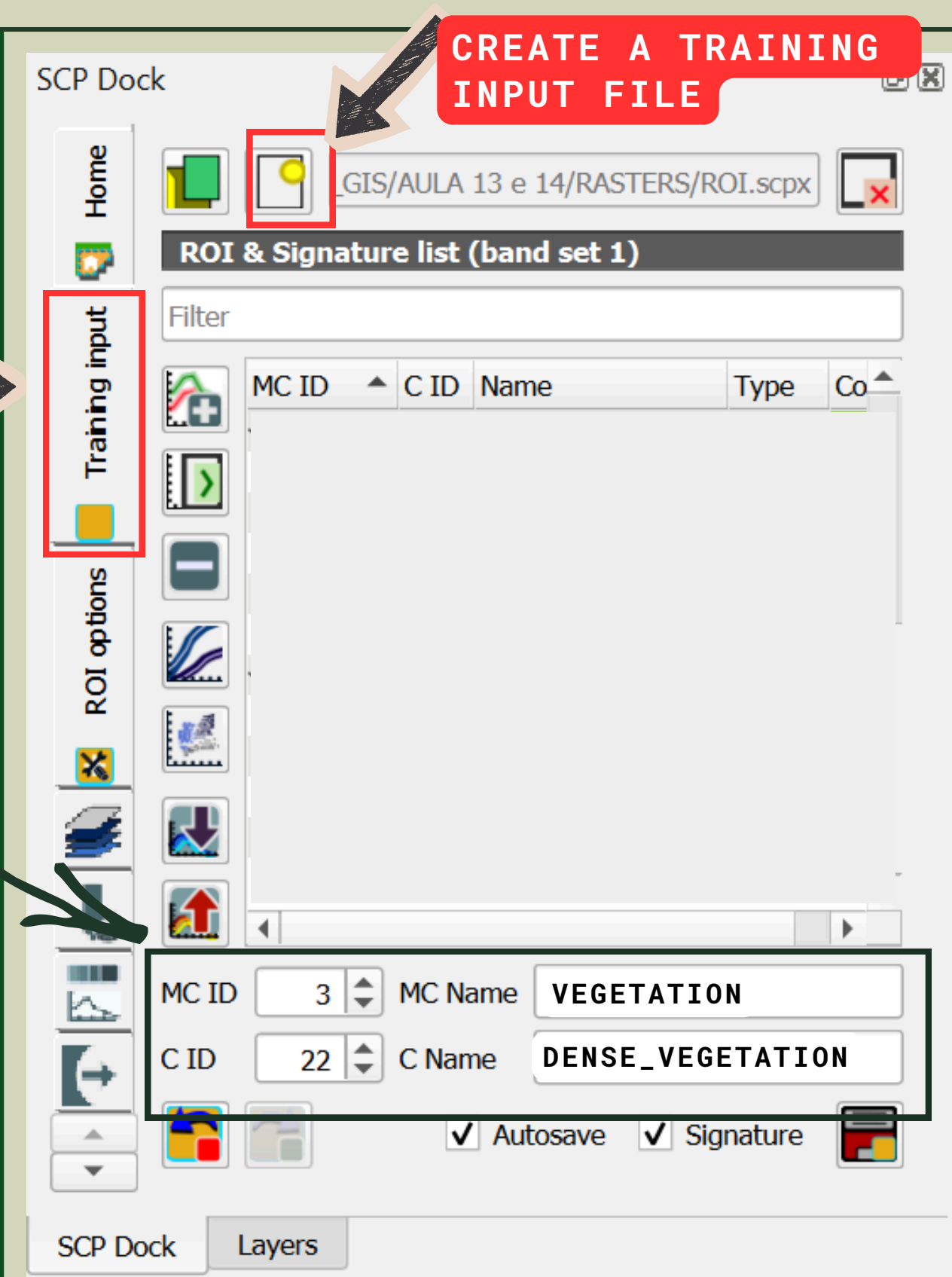
Band	Name	<u>Wavelength (nm)</u>
B2	Blue	<u>490</u>
B3	Green	<u>560</u>
B4	Red	<u>665</u>
B5	Red-edge 1	<u>705</u>
B6	Red-edge 2	<u>740</u>
B7	Red-edge 3	<u>783</u>
B8	NIR	<u>842</u>
B8A	Narrow NIR	<u>865</u>
B11	SWIR-1	<u>1610</u>
B12	SWIR-2	<u>2190</u>

STEP 2: CREATE REGIONS OF INTEREST

- ROI

MENU → SCP → BASIC TOOLS → MULTIPLE ROI CREATION

1. GO TO 'TRAINING INPUT'
2. CREATE A TRAINING INPUT FILE
3. SET THE CLASSES AND MACRO CLASSES (MC)
4. CREATE ROI POLYGON →  
5. SAVE EACH ROI POLYGON → 

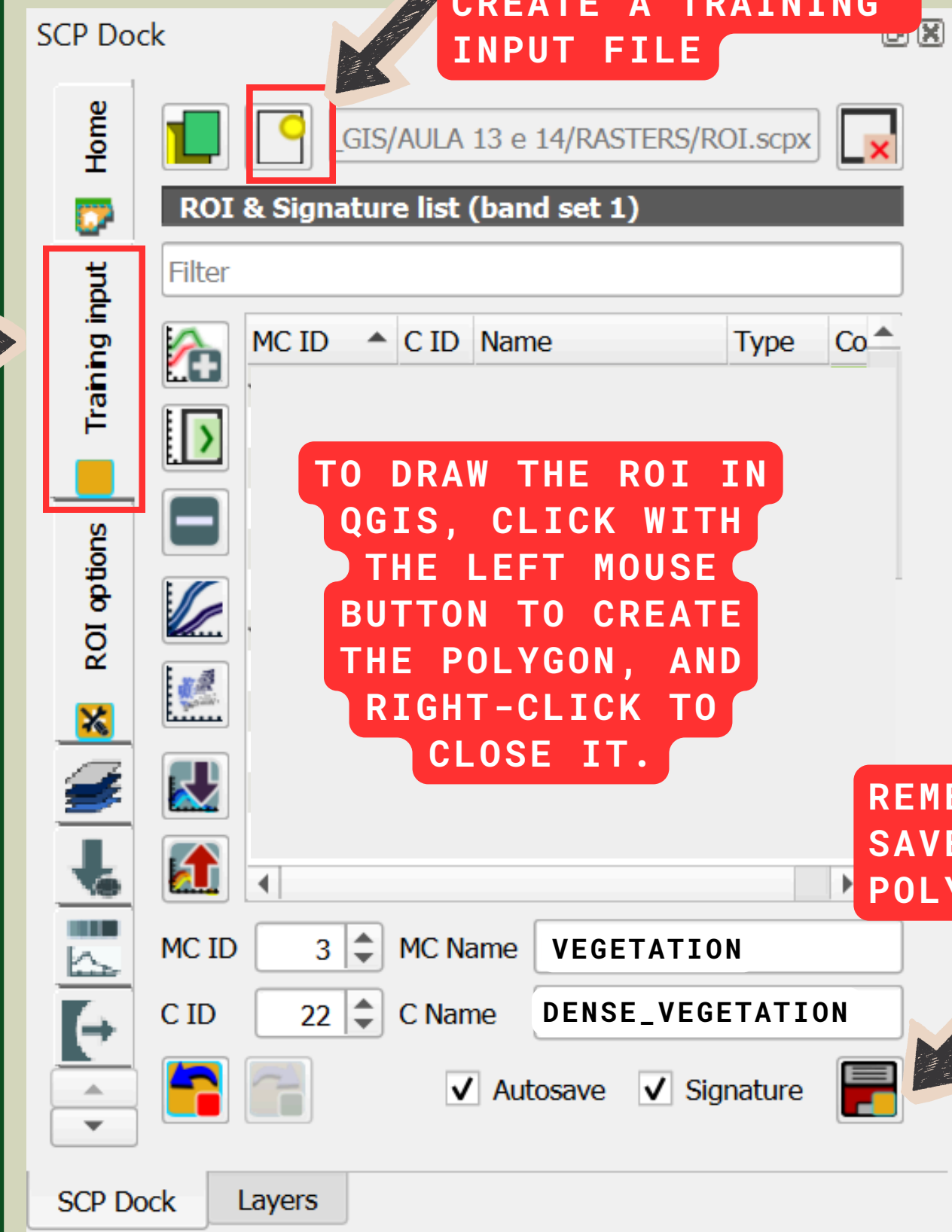


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SCP Dock

Home   GIS/AULA 13 e 14/RASTERS/ROI.scp 

ROI & Signature list (band set 1)

Filter

MC ID	C ID	Name	Type	Co
2 Water				
☒	2	11	Water	R&S
☒	2	13	Water	R&S
☒	2	14	Water	R&S
☒	2	15	Water	R&S
☒	2	16	Water	R&S
3 Build_up				
☒	3	17	Build_up	R&S
☒	3	18	Build_up	R&S
☒	3	19	Build_up	R&S
☒	3	20	Build_up	R&S
☒	3	21	Build_up	R&S

MC ID MC Name

C ID C Name

☒ Autosave ☒ Signature 

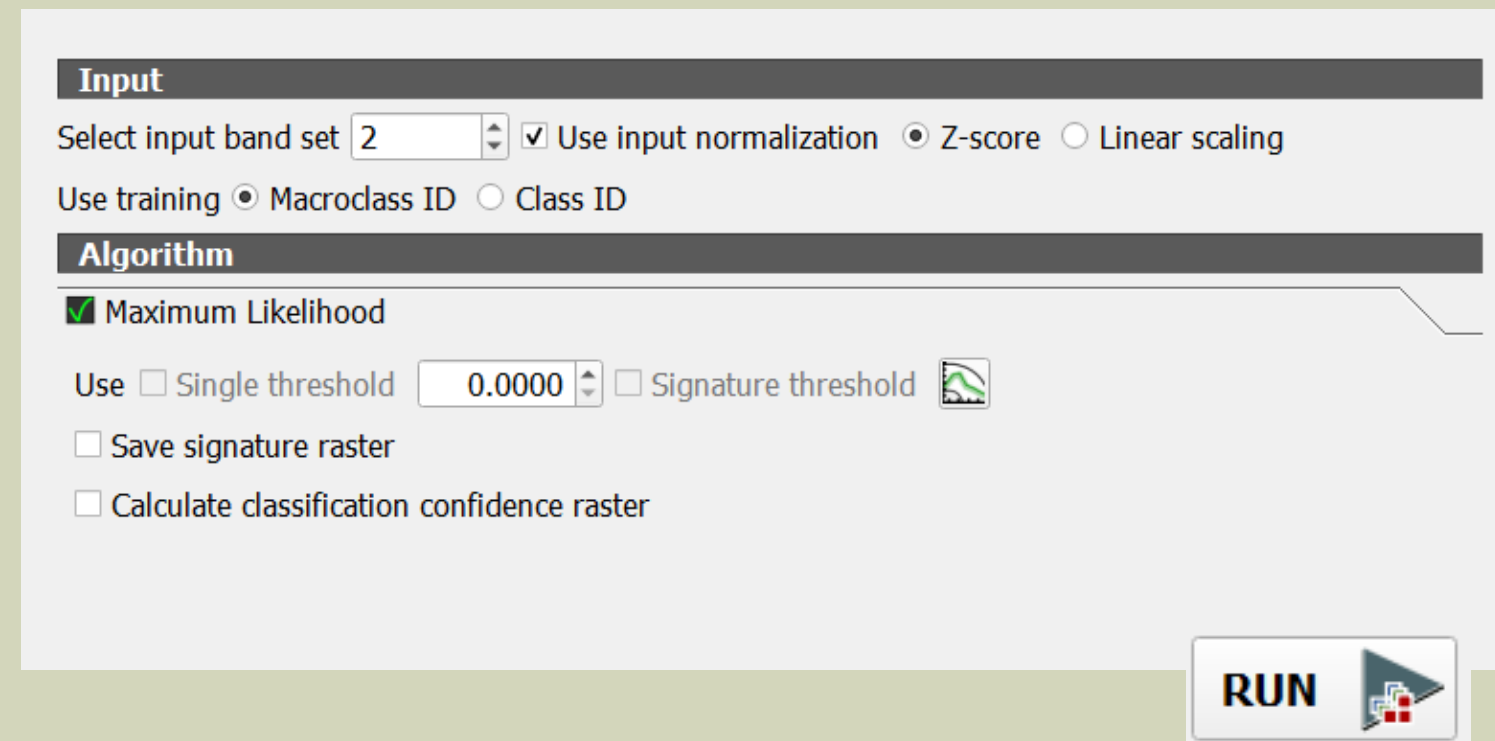
SCP Dock Layers

LIST OF
POLYGONS

STEP 3: RUN CLASSIFICATION

MENU → SCP → BAND PROCESSING → CLASSIFICATION

1. INPUT: SELECT INPUT BAND SET (IF YOU HAVE CREATED ONLY ONE BAND SET, "1" IS THE ONLY OPTION.
2. ALGORITHM: CHOOSE THE ALGORITHM
 - A. TIPS FOR MAXIMUM LIKELIHOOD:
 - I. CHECK Z-SCORE;
 - II. CHECK MACROCLASS ID
 - III. THRESHOLD: 0.05



The screenshot shows the SCP Classification tool interface. It has two main sections: 'Input' and 'Algorithm'. In the 'Input' section, 'Select input band set' is set to '2', 'Use input normalization' is checked, and 'Z-score' is selected for normalization. 'Use training' is set to 'Macroclass ID'. In the 'Algorithm' section, 'Maximum Likelihood' is selected. Under 'Use', 'Single threshold' is checked with a value of '0.0000', and 'Signature threshold' is unchecked. There are also unchecked checkboxes for 'Save signature raster' and 'Calculate classification confidence raster'. A 'RUN' button with a play icon is at the bottom right.

Input

Select input band set ☒ Use input normalization ☒ Z-score ☐ Linear scaling

Use training ☒ Macroclass ID ☐ Class ID

Algorithm

☒ Maximum Likelihood

Use ☐ Single threshold ☐ Signature threshold 

☐ Save signature raster

☐ Calculate classification confidence raster

RUN 